

Group -M

Feasibility Study

Project name: - STUDENT ATTENDANCE MANAGEMENT SYSTEM

Introduction

The Student Attendance Management System is designed to automate the attendance recording process using facial recognition technology. The system provides role-based access for administrators, teachers, and students while improving attendance accuracy, reducing manual workload, and enhancing record management. The feasibility study evaluates whether the project can be successfully implemented from technical, operational, financial, and scheduling perspectives.

1. Technical Feasibility

The proposed system is technically feasible because the required technologies are readily available, well-documented, and suitable for the project's requirements.

Technologies Used

- Python 3.x for application development
- CustomTkinter for modern graphical user interfaces
- MySQL Database for data storage
- OpenCV for facial recognition and image processing
- Face Recognition library for face detection and identification
- NumPy for facial encoding operations
- MySQL Connector for database connectivity
- Visual Studio Code for development and debugging

Technical Requirements

The system requires:

- A computer with a webcam
- Python development environment
- MySQL database server
- Face recognition libraries
- Sufficient storage for attendance records and facial data

Technical Challenges

Potential technical challenges include:

- Face recognition accuracy under different lighting conditions

- Managing large numbers of facial images
- Database optimization for faster attendance retrieval
- Integration between facial recognition and attendance modules

These challenges can be addressed through proper testing, image preprocessing, and database optimization.

Technical Feasibility Conclusion

The development team has access to all required technologies and development tools. Since all selected technologies are open-source and compatible with each other, the project is considered technically feasible.

2. Operational Feasibility

Operational feasibility evaluates whether the system will effectively solve the identified problem and be accepted by its intended users.

Current Problem

Traditional attendance systems:

- Require manual attendance marking
- Consume valuable classroom time
- Are prone to human errors
- Make attendance tracking difficult

Proposed Solution

The Student Attendance Management System provides:

- Automated facial recognition attendance
- Real-time attendance recording
- Attendance history tracking
- Attendance percentage calculations
- Report generation capabilities
- Multi-role access control for administrators, teachers, and students. Functional

User Benefits

Administrators

- Manage students and teachers
- Monitor attendance records
- Generate reports
- View attendance analytics

Teachers

- Mark attendance manually or automatically
- View attendance records
- Monitor absent students
- Access attendance reports

Students

- View personal attendance records
- Monitor attendance percentage
- Access attendance history

Operational Feasibility Conclusion

The proposed system significantly improves attendance management efficiency and provides clear benefits to all stakeholders. Therefore, the system is operationally feasible.

3. Financial Feasibility

Financial feasibility assesses the cost of developing and operating the system.

Software Costs

Software/Tool	Cost
Python	Free
OpenCV	Free
Face Recognition Library	Free
CustomTkinter	Free
MySQL	Free
Visual Studio Code	Free
GitHub	Free

Hardware Costs

Hardware	Cost
Computer/Laptop	Existing Resource
Webcam	Existing Resource or Low Cost

Development Cost

The project is developed by students using freely available resources and software tools, minimising development expenses.

Financial Feasibility Conclusion

Since all software technologies are open-source and the hardware requirements are minimal, the project can be completed with little to no additional cost. Therefore, the project is financially feasible.

4. Schedule Feasibility

Schedule feasibility determines whether the project can be completed within the available timeframe.

Development Plan

Sprint 1 (Weeks 1–2)

- Requirements gathering
- System architecture design
- Database design
- User authentication system
- User role implementation

Sprint 2 (Weeks 3–4)

- Student management
- Teacher management
- Attendance management
- Dashboard development

Sprint 3 (Weeks 5–6)

- Facial recognition attendance
- Analytics dashboard
- Report generation
- Testing and bug fixing
- Final deployment preparation

Risk Management

Potential schedule risks include:

- Face recognition implementation challenges
- Database integration issues

- Unexpected software bugs

These risks will be managed through Agile development practices, sprint reviews, and continuous testing.

Schedule Feasibility Conclusion

The project scope is realistic and achievable within the planned development period using Agile methodology. Therefore, the project is schedule feasible.

Overall Feasibility Assessment

The Student Attendance Management System is technically, operationally, financially, and schedule-wise feasible. The proposed solution uses accessible technologies, requires minimal cost, addresses a real-world educational problem, and can be completed within the allocated project timeframe. The system is therefore considered a viable and practical solution for automating student attendance management.

Feasibility Summary

Feasibility Type	Result
Technical Feasibility	Feasible
Operational Feasibility	Feasible
Financial Feasibility	Feasible
Schedule Feasibility	Feasible
Overall Project Viability	Highly Feasible